ANSWER KEY & MARKING SCHEME SET 1

Chapter 6: **How Forces Affect Motion** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
2	**(c)** 1 : 4	a is inversely proportional to m.
3	**(d)** the pull of the book on the Earth	Third-law pair.
4	**(d)** 4 m/s squared	F/m.
5	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
6	**(d)** 20 m/s	a = 4 m/s squared; v = 20 m/s.
7	**(b)** reducing the friction by rollers	Reduce friction.
8	**(a)** 200 N	0.1 x 20 / 0.01.
9	**(a)** Both A and R are true and R is the correct explanation of A.	This is inertia of motion, which is the reason for the forward jerk.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	Both are true, but they are unrelated statements.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	a = -2 m/s squared (1); t = 5 s, s = 25 m (1).	1+1
12	Normal force from the ground balances the weight (1); Net force zero (1).	1+1
13	Net force zero at constant velocity (1); Hence pull equals friction (1).	1+1
14	Same direction: 14/2 = 7 m/s squared (1); Opposite: 4/2 = 2 m/s squared (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	a = 4 m/s squared (1); F = 6000 N, s = 50 m (1); Weight = 15000 N; force is 0.4 of weight (1).	1+1+1
16	a = v squared/2s = 2 x 10^6 m/s squared (1); F = 0.01 x 2 x 10^6 = 20000 N (1); t = v/a = 0.0002 s (1).	1+1+1
17	F = 10/0.01 = 1000 N (1); Third law: nail exerts 1000 N on the hammer (1); Force on the nail 1000 N (1).	1+1+1
	OR: a = 5 m/s squared (1); v = 20 m/s, change in momentum 120 kg m/s (1); s = 40 m (1).	1+1+1
18	At 40 N: net force zero, constant velocity or rest (1); At 60 N: net 20 N (1); Box accelerates in the direction of the push (1).	1+1+1
	OR: To every action there is an equal and opposite reaction (1); Examples: recoil of a gun, rocket, walking (1); Action and reaction act on different bodies (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $W = mg = 20 \times 10 = 200$ N (1). (ii) $50 - 10 = 40$ N (1). (iii) $a = F/m = 40/20 = 2$ m/s^2 (1). (iv) $v = at = 2 \times 5 = 10$ m/s (1); $s = \frac{1}{2}at^2 = \frac{1}{2} \times 2 \times 25 = 25$ m (1).	1+1+1+2
20	Momentum before firing = 0 (1); Momentum of bullet = 0.02 x 200 = 4 kg m/s (1); Gun has equal momentum in the opposite direction (1); Recoil velocity = 4/4 = 1 m/s opposite to the bullet (1); Law of conservation of momentum (1).	1+1+1+1+1
	OR: Net = sqrt(900 + 1600) = 50 N (1); a = 10 m/s squared (1); tan = 40/30, about 53 degrees north of east (1); Second law (1); Vector addition (1).	1+1+1+1+1

Mistakes to avoid:

- Using mass in kg as weight in N (a 5 kg body weighs 50 N when $g = 10$), or saying that the mass of a body changes on the Moon.
- Using the applied force instead of the *net* force in $F = ma$ when friction acts.
- Saying that action and reaction cancel; they are equal and opposite but act on two different bodies.
- Forgetting to convert grams to kilograms (a 20 g bullet is 0.02 kg) in momentum and recoil problems.

- Dropping the negative sign (direction) of recoil velocity, or believing that a heavier gun recoils faster; recoil speed is inversely proportional to the gun's mass.
- Confusing inertia of rest with inertia of motion, or thinking that inertia depends on speed; it depends only on mass.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 2

Chapter 6: **How Forces Affect Motion** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** the same for both	F = change in p/t.
2	**(c)** 10 N	20/2.
3	**(c)** A is true but R is false.	A is true; R is false: mass does not depend on gravity.
4	**(c)** 300 N	m a + f = 200 + 100.
5	**(b)** 1 : 2	v = Ft/m.
6	**(b)** 10 N s	Ft.
7	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
8	**(c)** 6 N	a = 2; force on the 3 kg block = 6.
9	**(c)** A is true but R is false.	$W = mg$, so weight depends on g; R is false.
10	**(d)** A is false but R is true.	Forces can cancel only if they act on the same body, so A is false; R is true.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	a = F/m = 40/8 (1); = 5 m/s squared (1).	1+1
12	Impulse 1 N s (1); v = 2 m/s (1).	1+1
13	F = rate of change of momentum (1); = 5 x 2000 = 10000 N (1).	1+1
14	Body keeps the forward velocity of the train (1); Feet stop on the ground, upper body continues and he falls (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	a = F/m = 60/12 = 5 m/s squared (1); v = at = 5 x 3 (1); = 15 m/s (1).	1+1+1
16	Limiting static friction = 250 N (1); Net force when moving = 100 N (1); a = 2 m/s squared (1).	1+1+1
17	a = 3 m/s squared (1); v = 15 m/s (1); From rest (1).	1+1+1
	OR: a = 5 m/s squared for 2 s: v = 10 m/s (1); Then constant 10 m/s (1); s = 10 + 30 = 40 m (1).	1+1+1
18	Same bullet momentum (1); Recoil speed inversely proportional to mass (1); Heavier gun recoils less (ratio 4 : 1) (1).	1+1+1
	OR: a = 2 m/s squared (1); v = 8 m/s (1); Change in momentum = 12 kg m/s (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $p = mv = 0.15 \times 30 = 4.5$ kg m/s (1). (ii) $F = \frac{4.5 - 0}{0.05} = 90$ N (1). (iii) $F = \frac{4.5}{0.15} = 30$ N (1). (iv) Newton's second law, $F = \Delta p / \Delta t$ (1); the change of momentum is the same, but a longer time gives a smaller rate of change, hence a smaller force (1).	1+1+1+2
20	Change in momentum 900 kg m/s for passenger (1); With airbag (0.3 s): 3000 N (1); Without (0.1 s): 9000 N (1); Airbag reduces the force to one-third (1); F = change in momentum/time (1).	1+1+1+1+1
	OR: Engine force overcomes resistance (1); Net force gives acceleration (1); Inertia in braking (1); Seat belt, airbag, crumple zone (1); Speed limits reduce momentum (1).	1+1+1+1+1

Mistakes to avoid:

- Using mass in kg as weight in N (a 5 kg body weighs 50 N when $g = 10$), or saying that the mass of a body changes on the Moon.
- Using the applied force instead of the *net* force in $F = ma$ when friction acts.
- Saying that action and reaction cancel; they are equal and opposite but act on two different bodies.
- Forgetting to convert grams to kilograms (a 20 g bullet is 0.02 kg) in momentum and recoil problems.
- Dropping the negative sign (direction) of recoil velocity, or believing that a heavier gun recoils faster; recoil speed is inversely proportional to the gun's mass.
- Confusing inertia of rest with inertia of motion, or thinking that inertia depends on speed; it depends only on mass.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 3

Chapter 6: **How Forces Affect Motion** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
2	**(d)** moves away from the shore	Third law.
3	**(a)** 500 N	m a on the trailer.
4	**(a)** 4 m	v squared/2a.
5	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
6	**(d)** 7 m/s	a = 2; v = 3 + 4.
7	**(d)** a1 a2/(a1 + a2)	F/(m1 + m2) with m = F/a.
8	**(b)** 10 m/s	a = 2.5.
9	**(a)** Both A and R are true and R is the correct explanation of A.	Conservation of momentum explains the recoil.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	Both are true; R does not explain why rolling friction is smaller.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Friction at the axle and air drag (1); Unbalanced force slows it (1).	1+1
12	v = 10 m/s (1); F = 600/0.2 = 3000 N (1).	1+1
13	a = -2 m/s squared; F = 4 N (1); s = 9 m (1).	1+1
14	100 N in the opposite direction (1); Wall does not move due to ground friction (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Air resistance increases with speed (1); At terminal velocity net force zero (1); Resistance = weight = 700 N (1).	1+1+1
16	Change = 0.15 x 70 = 10.5 kg m/s (1); F = 1050 N (1); Impulse 10.5 N s (1).	1+1+1
17	a1 = 6 m/s squared (1); a2 = 2 m/s squared (1); More inertia, less acceleration (1).	1+1+1
	OR: (a) 50 N (1); (b) 60 N (1); (c) 40 N (1).	1+1+1
18	The stone moves along the tangent (1); No force acts to change its direction (1); Inertia of motion (1).	1+1+1
	OR: F = 3 N (1); u = 4 m/s (1); v = 8 m/s (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Zero (both at rest) (1). (ii) $0.02 \times 200 = 4$ kg m/s (1). (iii) By conservation of momentum, $4 \times v + 4 = 0$, so $v = -1$ m/s, i.e. 1 m/s (1). (iv) The momenta are equal in size and the rifle has a much larger mass, so $v = p/m$ is small (1); the rifle moves opposite to the bullet (1).	1+1+1+2
20	Net force 40 N (1); a = 4 m/s squared (1); v = 20 m/s (1); s = 50 m (1); p = 200 kg m/s (1).	1+1+1+1+1
	OR: a = 5 m/s squared (1); s = 40 m, t = 4 s (1); Double speed: s = 160 m, t = 8 s (1); Distance four times, time twice (1); Important for speed limits (1).	1+1+1+1+1

Mistakes to avoid:

- Using mass in kg as weight in N (a 5 kg body weighs 50 N when $g = 10$), or saying that the mass of a body changes on the Moon.
- Using the applied force instead of the *net* force in $F = ma$ when friction acts.
- Saying that action and reaction cancel; they are equal and opposite but act on two different bodies.
- Forgetting to convert grams to kilograms (a 20 g bullet is 0.02 kg) in momentum and recoil problems.
- Dropping the negative sign (direction) of recoil velocity, or believing that a heavier gun recoils faster; recoil speed is inversely proportional to the gun's mass.

- Confusing inertia of rest with inertia of motion, or thinking that inertia depends on speed; it depends only on mass.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND